

Created Feb 10Feb 10

```
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#####

## update the system packages
sudo apt-get update

## #1: Install OpenJDK 17 on Ubuntu 24.04 LTS
apt install openjdk-17-jdk openjdk-17-jre -y

##Download the SonaType Nexus on Ubuntu using wget
cd /opt

sudo wget https://download.sonatype.com/nexus/3/latest-unix.tar.gz

##Extract the Nexus repository setup in /opt directory
tar -zxvf latest-unix.tar.gz

##Rename the extracted Nexus setup folder to nexus
sudo mv /opt/nexus-3.76.1-01/ /opt/nexus

##As security practice, not to run nexus service using root user, so lets create
new user named nexus to run nexus service
sudo adduser nexus --> give password

##To set no password for nexus user open the visudo file in ubuntu
sudo visudo

##Add below line into it , save and exit
nexus ALL=(ALL) NOPASSWD: ALL

##Give permission to nexus files and nexus directory to nexus user

sudo chown -R nexus:nexus /opt/nexus
sudo chown -R nexus:nexus /opt/sonatype-work

##To run nexus as service at boot time, open /opt/nexus/bin/nexus.rc file,
uncomment it and add nexus user as shown below

sudo vi /opt/nexus/bin/nexus.rc

run_as_user="nexus"

##To Increase the nexus JVM heap size, you can modify the size as shown below

vi /opt/nexus/bin/nexus.vmoptions

-Xms1024m
-Xmx1024m
-XX:MaxDirectMemorySize=1024m

-XX:LogFile=./sonatype-work/nexus3/log/jvm.log
-XX:-OmitStackTraceInFastThrow
-Djava.net.preferIPv4Stack=true
-Dkaraf.home=.
-Dkaraf.base=.
-Dkaraf.etc=etc/karaf
-Djava.util.logging.config.file=/etc/karaf/java.util.logging.properties
```

```
-Dkaraf.data=./sonatype-work/nexus3
-Dkaraf.log=./sonatype-work/nexus3/log
-Djava.io.tmpdir=./sonatype-work/nexus3/tmp
```

##To run nexus as service using Systemd

```
sudo vi /etc/systemd/system/nexus.service
```

```
[Unit]
```

```
Description=nexus service
```

```
After=network.target
```

```
[Service]
```

```
Type=forking
```

```
LimitNOFILE=65536
```

```
ExecStart=/opt/nexus/bin/nexus start
```

```
ExecStop=/opt/nexus/bin/nexus stop
```

```
User=nexus
```

```
Restart=on-abort
```

```
[Install]
```

```
WantedBy=multi-user.target
```

```
-----
```

```
sudo systemctl start nexus
```

```
sudo systemctl enable nexus
```

```
sudo systemctl status nexus
```

```
tail -f /opt/sonatype-work/nexus3/log/nexus.log
```

```
ufw allow 8081/tcp --> if required
```

```
http://server_IP:8081
```

```
=====
```

```
Click on SignIN --> Username = admin, password = cat
/opt/nexus/sonatype-work/nexus3/admin.password
```

```
Creating Repo
```

```
=====
```

```
Click on Setting Symbol --> Repositories --> Create repository -->
maven2(hosted) --> name(hotstar) --> save
```

```
Version Policy --> Snapshot
```

```
Deployment policy --> allow to redeploy
```

```
Integrate Nexus to Jenkins Pipeline
```

```
=====
```

```
Jenkins (Code --> Build --> Test --> Artifact) --> Nexus
```

```
If you want to integrate any 3rd party to Jenkins, install plugins
```

```
1. Download the Plugins (nexus artifacts uploader)
--> Manage Jenkins --> Plugins --> Available Plugins --> Just type only Nexus
, it will give nexus artifacts uploader
```

2. Configure in Jenkins pipeline in stage

Note: all the information will be available in pom.xml file

First create Credentials for Nexus

Manage Jenkins --> Credentials --> Username = admin, password = root123 , id = nexus , description = nexus

Create a new pipeline job --> and copy paste the script and click on pipeline syntax

sample step : nexusartifactuploader

Nexus Version : see nexus version in browser left top 3.0

Protocol:http

Nexus URL : IP:8081 (dont put http)

Credentials : Add --> Jenkins --> username:admin, password: root123, description: nexus

credentials: select admin

Groupid: see pom file (in.reyaz)

version: see pom file (1.2.2) 8.3.3-SNAPSHOT

Repository: repo that you created in nexus : hotstar

Artifact: Add : Artifactid: see pom (myapp) , Type = .war, classifier = empty, File = target/myapp.war

--> generate pipeline script

```
nexusArtifactUploader artifacts: [[artifactId: 'myapp', classifier: '', file:
'target/myapp.war', type: '.war']], credentialsId: 'nexus', groupId: 'in.reyaz',
nexusUrl: '13.126.193.100:8081', nexusVersion: 'nexus3', protocol: 'http',
repository: 'hotstar', version: '8.3.3-SNAPSHOT'
```

```
pipeline {
    agent any

    stages {
        stage('checkout') {
            steps {
                git 'https://github.com/ReyazShaik/java-project-maven-
new.git#39;
            }
        }
        stage('compile') {
            steps {
                sh 'mvn compile'
            }
        }
        stage('test') {
            steps {
                sh 'mvn test'
            }
        }
        stage('artifact') {
            steps {
                sh 'mvn clean package'
            }
        }
        stage('nexus') {
            steps {
                nexusArtifactUploader artifacts: [[artifactId: 'myapp',
classifier: '', file: 'target/myapp.war', type: '.war']], credentialsId:
'nexuscreds', groupId: 'in.reyaz', nexusUrl: '13.127.129.23:8081', nexusVersion:
'nexus3', protocol: 'http', repository: 'hotstarapp', version: '8.3.3-SNAPSHOT'
            }
        }
    }
}
```

}

}

}